impossible (sorry). 3) As something approaches absolute zero temperature, its entropy approaches a constant value.

10 This isn't a tangible surface, but rather a sphere in space defined by the distance from the black hole's center to the Schwarzschild radius, which is what we call the distance from the singularity out to the horizon. The Schwarzschild radius is directly related to the black hole's mass.

11 Real particles can't have negative energy, but these are virtual particles, which are just a totally different kind of beast, and are not to be confused with negatively *charged* particles like electrons.

12 You may recognize Dyson's name from the sci-fi concept of a "Dyson sphere"—a monumentally huge sphere built around a star to capture 100 percent of its radiation for the purposes of powering an advanced alien civilization. Observational surveys for Dyson spheres, which look for the waste heat expected to be emitted by them in the infrared, have so far come up empty.

5. Big Rip

1 He has to be wrong about *something*.

2 In explaining the adoption of the term "phantom" in the first paper to touch on this idea in 1999, Caldwell wrote, "A phantom is something which is apparent to the sight or other senses but has no corporeal existence—an appropriate description for a form of energy necessarily described by unorthodox physics."

3 The newest one, called Gaia, is producing spectacularly detailed maps of the stars in our galaxy and is already giving us incredible insights into our cosmic history. What it will tell us about our fate is to be determined.

4 When the danger is space itself, you want to be in a structure that has as little space in it as possible.

5 If you ask them, my colleagues will claim that their real motivation is understanding the nature of dark energy because of